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Summary

Full-Stack Software Engineer specializing in legacy system modernization and high-performance internal tooling. Proven track record of decoupling monolithic architectures and scaling enterprise tools for 500+ enterprise users.

IIIT Trichy | Bachelors of Technology (B.Tech.) | 2020-2024

Technical Skills

- **Languages** — C/C++, Java, TypeScript, JavaScript, Python
- **Frontend** — ReactJS, NextJS, ThreeJS, WebGL, Shaders, REST APIs, JWT/Auth 2.0, WebSockets
- **Backend & APIs** — NodeJS, ExpressJS, Flask
- **Databases** — MongoDB, PostgreSQL, Procedural Language SQL, Firebase Console
- **Cloud Integrations & DevOps** — AWS (EC2/ Lambda), Cloudflare, Git, RazorPay
- **AI Tools** — OpenAI API, Anthropic Claude, Prompt Engineering, AI-assisted Development

Experience

Systems Engineer — TCS (Nov 2024 – Present)

- **Legacy to Modern API Migration:** Orchestrated the migration of legacy **SOAP-based architecture to scalable REST APIs, decoupling core transaction logic** to modernize the enterprise middleware stack.
- **Backend Performance Optimization:** Engineered asynchronous, **multi-threaded callback logic** that eliminated redundant operations and **reduced database query latency from 5+ seconds to sub-second** response times.
- **Enterprise Frontend & Tooling:** Architected and led the **end-to-end full-stack delivery** of 'PDN Tracker', building a highly responsive **React.js UI** to replace a deprecated, security-compromised vendor tool.
- **Production Scale:** Successfully drove the tracker to production, managing **role-based access control (RBAC)** and deployment pipelines for **500+ daily active enterprise users**.

Projects

PDN Tracker - Enterprise Deployment Management System | ReactJS, TailwindCSS, NodeJS, PostgreSQL/SpringBoot

- **Rapid Lifecycle Development:** Co-developed a JIRA-style tracking replacement in under 3 months, engineering a robust **state machine** to track complex deployment lifecycles
- **Frontend & UI Architecture:** Solely owned the **React.js UI**, implementing **session-based authentication**, dynamic role-specific dashboards (Developer Workspaces vs. Reviewer Queues), and a global indexing search for instant PDN retrieval.
- **Backend Node.js Pivot:** Independently re-architected the legacy Java backend into a modern **Node.js, Express, and MongoDB REST API** to make the platform independently hostable and highly scalable.
- **Business Impact:** Engineered strict **Role-Based Access Control (RBAC)** workflows to ensure secure cross-team handoffs; system is currently active in production and utilized by **500+ enterprise users daily**.

Chunky Discord — Zero-Cost Distributed File Storage | ReactJS, NodeJS, Firebase DB, AWS EC2

- **System Architecture:** Architected a distributed cloud storage system to bypass traditional cloud fees by splitting large files into manageable chunks and routing them through **Discord server APIs** as a zero-cost backend.
- **Data Integrity:** Engineered a robust **hash-based retrieval and reassembly algorithm** that guarantees 100% file integrity upon download, ensuring the host server remains entirely stateless and stores zero payload data.
- **Cloud Deployment:** Deployed the highly scalable architecture on **AWS EC2** with secure **Firebase authentication**, demonstrating advanced system design and cost-optimized engineering.

Competitive Programming

LeetCode rating 1834 — 500+ problems solved across Data Structures and algorithmic challenges